

X	Y
1	2
2	3
3	5

1) Mean (vector)

$$\mu = \begin{bmatrix} \bar{x} \\ \bar{y} \end{bmatrix} = \begin{bmatrix} (1+2+3)/3 \\ (2+3+5)/3 \end{bmatrix} = \begin{bmatrix} 2 \\ 10/3 \end{bmatrix}.$$

2) Covariance via dot products (linear algebra)

Form the **centered data** (subtract the mean from each point):

$$Z = \begin{bmatrix} -1 & -4/3 \\ 0 & -1/3 \\ 1 & 5/3 \end{bmatrix} \quad (\text{rows are centered samples}).$$

Let z_1 be the first column of Z and z_2 the second:

$$z_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad z_2 = \begin{bmatrix} -4/3 \\ -1/3 \\ 5/3 \end{bmatrix}.$$

The (population) covariance matrix is

$$\Sigma = \frac{1}{n} Z^\top Z = \frac{1}{3} \begin{bmatrix} z_1 \cdot z_1 & z_1 \cdot z_2 \\ z_2 \cdot z_1 & z_2 \cdot z_2 \end{bmatrix}.$$

Compute the dot products:

- $z_1 \cdot z_1 = (-1)^2 + 0^2 + 1^2 = 2.$
- $z_2 \cdot z_2 = \frac{16}{9} + \frac{1}{9} + \frac{25}{9} = \frac{42}{9} = \frac{14}{3}.$
- $z_1 \cdot z_2 = (-1)(-4/3) + 0(-1/3) + 1(5/3) = \frac{9}{3} = 3.$

Thus

$$\Sigma = \frac{1}{3} \begin{bmatrix} 2 & 3 \\ 3 & 14/3 \end{bmatrix} = \begin{bmatrix} 2/3 & 1 \\ 1 & 14/9 \end{bmatrix}.$$

(Notice the **non-zero off-diagonal** = 1, so features are correlated.)

For later, its determinant and inverse:

$$\det(\Sigma) = \frac{2}{3} \cdot \frac{14}{9} - 1 = \frac{28}{27} - 1 = \frac{1}{27},$$

$$\Sigma^{-1} = \frac{1}{\det(\Sigma)} \begin{bmatrix} 14/9 & -1 \\ -1 & 2/3 \end{bmatrix} = 27 \begin{bmatrix} 14/9 & -1 \\ -1 & 2/3 \end{bmatrix} = \begin{bmatrix} 42 & -27 \\ -27 & 18 \end{bmatrix}.$$

3) Multivariate normal PDF at a test point

Take $x = \begin{bmatrix} 2.5 \\ 4 \end{bmatrix}$.

$$\delta = x - \mu = \begin{bmatrix} 2.5 - 2 \\ 4 - 10/3 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 2/3 \end{bmatrix}.$$

Quadratic form (pure linear algebra):

$$Q = (x - \mu)^\top \Sigma^{-1} (x - \mu) = \delta^\top \Sigma^{-1} \delta.$$

Compute $u = \Sigma^{-1} \delta$:

$$u = \begin{bmatrix} 42 & -27 \\ -27 & 18 \end{bmatrix} \begin{bmatrix} 1/2 \\ 2/3 \end{bmatrix} = \begin{bmatrix} 42 \cdot \frac{1}{2} - 27 \cdot \frac{2}{3} \\ -27 \cdot \frac{1}{2} + 18 \cdot \frac{2}{3} \end{bmatrix} = \begin{bmatrix} 21 - 18 \\ -13.5 + 12 \end{bmatrix} = \begin{bmatrix} 3 \\ -3/2 \end{bmatrix}.$$

Then

$$Q = \delta^\top u = \frac{1}{2} \cdot 3 + \frac{2}{3} \cdot \left(-\frac{3}{2}\right) = \frac{3}{2} - 1 = \frac{1}{2}.$$

Finally, the 2-D Gaussian PDF:

$$f(x) = \frac{1}{(2\pi)^{d/2} \sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}Q\right) = \frac{1}{(2\pi)^1 \sqrt{1/27}} \exp\left(-\frac{1}{2} \cdot \frac{1}{2}\right).$$

Since $\sqrt{1/27} = 1/(3\sqrt{3})$,

$$f(x) = \frac{3\sqrt{3}}{2\pi} e^{-1/4} \approx 0.644.$$

Orthogonality link

- If Z_1 and Z_2 are **orthogonal vectors**, their dot product = 0.
- That means:

$$Z_1 \cdot Z_2 = 0 \implies \text{Cov}(X_1, X_2 \mid c_i) = 0$$

- Geometrically: the variations of X_1 and X_2 are “independent directions” — no linear correlation.

- In the **full Bayes classifier**, each class c_i is modeled with a **multivariate Gaussian**:

$$P(x|c_i) = \frac{1}{(2\pi)^{d/2} |\Sigma_i|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_i)^T \Sigma_i^{-1} (x - \mu_i)\right)$$

Here Σ_i is the **covariance matrix** of class i . Normally, Σ_i contains variances (diagonal terms) and covariances (off-diagonal terms).

- But in **Naive Bayes**, the assumption is that features are independent:

$$P(x|c_i) = \prod_{j=1}^d P(x_j|c_i)$$

That assumption is equivalent to **setting all off-diagonal entries of Σ_i to zero**.

- Bayes theorem:

$$P(c_i|x) \propto P(x|c_i) P(c_i)$$

- In **Naive Bayes**, we assume independence across features:

$$P(x|c_i) = \prod_{j=1}^d P(x_j|c_i)$$

For **numeric attributes**, we assume each feature follows a **Gaussian distribution**:

$$P(x_j|c_i) = \frac{1}{\sqrt{2\pi} \sigma_{ij}} \exp\left(-\frac{(x_j - \mu_{ij})^2}{2\sigma_{ij}^2}\right)$$

This is equivalent to assuming the covariance matrix Σ_i is **diagonal**:

$$\Sigma_i = \begin{bmatrix} \sigma_{i1}^2 & 0 & \cdots & 0 \\ 0 & \sigma_{i2}^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{id}^2 \end{bmatrix}$$

So instead of:

$$\Sigma_i = \begin{bmatrix} \sigma_{i1}^2 & \text{Cov}(X_1, X_2) & \dots \\ \text{Cov}(X_2, X_1) & \sigma_{i2}^2 & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

we use a **diagonal covariance matrix**:

$$\Sigma_i = \begin{bmatrix} \sigma_{i1}^2 & 0 & \dots & 0 \\ 0 & \sigma_{i2}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{id}^2 \end{bmatrix}$$

Suppose we have two classes: $c_1 = \text{"Class A"}$, $c_2 = \text{"Class B"}$.

Features: X_1 (height), X_2 (weight).

Training data:

Class	X1 (height)	X2 (weight)
A	5.0	50
A	5.5	52
A	6.0	49
B	5.0	70
B	5.5	68
B	6.0	72

3. Step 1 – Priors

Each class has 3 samples, total = 6.

$$P(c_1) = \frac{3}{6} = 0.5, \quad P(c_2) = \frac{3}{6} = 0.5$$

4. Step 2 – Means and Diagonal Covariance

For **Class A**:

- Means:

$$\mu_{A1} = \frac{5.0 + 5.5 + 6.0}{3} = 5.5, \quad \mu_{A2} = \frac{50 + 52 + 49}{3} = 50.33$$

- Variances:

$$\sigma_{A1}^2 = \frac{(5.0 - 5.5)^2 + (5.5 - 5.5)^2 + (6.0 - 5.5)^2}{3} = 0.167$$
$$\sigma_{A2}^2 = \frac{(50 - 50.33)^2 + (52 - 50.33)^2 + (49 - 50.33)^2}{3} = 1.556$$

So covariance matrix:

$$\Sigma_A = \begin{bmatrix} 0.167 & 0 \\ 0 & 1.556 \end{bmatrix}$$

For **Class B**:

- Means:

$$\mu_{B1} = 5.5, \quad \mu_{B2} = \frac{70 + 68 + 72}{3} = 70$$

- Variances:

$$\sigma_{B1}^2 = 0.167, \quad \sigma_{B2}^2 = 2.667$$

So covariance matrix:

$$\Sigma_B = \begin{bmatrix} 0.167 & 0 \\ 0 & 2.667 \end{bmatrix}$$

5. Step 3 – Classifying a New Point

Say we test with $x = (5.4, 51)$.

For Class A:

$$\begin{aligned}P(x_1|A) &= \frac{1}{\sqrt{2\pi \cdot 0.167}} \exp\left(-\frac{(5.4 - 5.5)^2}{2 \cdot 0.167}\right) \\&= 0.977 \\P(x_2|A) &= \frac{1}{\sqrt{2\pi \cdot 1.556}} \exp\left(-\frac{(51 - 50.33)^2}{2 \cdot 1.556}\right) \\&= 0.316\end{aligned}$$

So:

$$P(x|A) \approx 0.977 \times 0.316 = 0.309$$

Posterior:

$$P(A|x) \propto 0.309 \times 0.5 = 0.154$$

For Class B:

$$\begin{aligned}P(x_1|B) &= \frac{1}{\sqrt{2\pi \cdot 0.167}} \exp\left(-\frac{(5.4 - 5.5)^2}{2 \cdot 0.167}\right) = 0.977 \\P(x_2|B) &= \frac{1}{\sqrt{2\pi \cdot 2.667}} \exp\left(-\frac{(51 - 70)^2}{2 \cdot 2.667}\right) \approx 0 \text{ (very small)}\end{aligned}$$

So:

$$P(x|B) \approx 0.977 \times 0 = 0$$

Posterior:

$$P(B|x) \approx 0$$

